Image Processing – HW1

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Please refer to the README.md first, the following only explain the code logic and result.

Q1:

1. We need to find the most top pixel of the 2 image, white pixel for lena_left and non-white pixel for lena_cropped.
2. We calculate the offset between these 2 coordinates so that we can paste the lena_cropped onto lena_left like a sticker.
3. When pasting, we need to mask out the white area in lena_cropped (background removal).
4. Done

The merged result:

Q2:

1. We use numpy's advance indexing to implement the resize() and rotate().
2. No matter if it's resizing or rotating, the steps are the same.
   a. Get the target image dimension
   b. Backward wrap every pixel in the target image (I named it src_x and src_y in the code)
   c. Apply interpolation strategy
   d. Create and return the output target image
3. For detail logic, refer to the code itself (I think I wrote the comments pretty detailed).
4. The one that is worth noting is the Bicibic Interpolation, I didn't use the method taught by the teacher in class. Instead, after doing some research, I found that using Keys' Cubic Convolution can be easier to implement (while keeping the same effect). I've attached the reference here (https://en.wikipedia.org/wiki/Bicubic_interpolation#Bicubic_convolution_algorithm), as you can see, we don't have to handle the derivatives and complex calculation together with advanced indexing.

Result:

Since it's about resizing, it's hard to show them here in PDF. Simple refer to the /result/q2 for more.

Note that "nni" stands for Nearest Neighbor Interpolation, "bi" stands for Bilinear Interpolation, and "bci" stands for Bicubic Interpolation.

https://github.com/bscny/college/tree/main/4th_grade/hw_image_processing-2/hw1/result/q2 is also available.

Q3:

1. Use painter to find the desired pixel, 4 source points and 4 corresponding target points. (blue is source, green is target)

2. Use cv2.getPerspectiveTransform to get a 3-by-3 matrix
3. Apply the cv2.warpPerspective

Result: